

Dear Hiring Committee,

I am excited to apply for the Research Assistant position in Signal Processing and Machine Learning at Brigham and Women's Hospital and Harvard Medical School. I am currently pursuing a Master of Science in Information Management at the University of Illinois Urbana-Champaign, where I conduct research involving machine learning, time-series analysis, and large-scale health-related datasets. The opportunity to develop AI solutions that directly improve maternal health outcomes is particularly motivating to me because it combines rigorous research with meaningful real-world impact.

In my current role as a Research Assistant at the University of Illinois, I design and implement machine learning pipelines for wearable sensor data collected from over 100 users. This work requires processing noisy physiological time-series data, handling missing values and irregular sampling, engineering meaningful features, and developing models that support longitudinal behavioral analysis. I have built Python-based workflows using NumPy, SciPy, PyTorch, and related tools to transform complex real-world datasets into reliable and reproducible analyses.

My research experience has also involved developing scalable data pipelines, performing signal preprocessing and pattern detection on large time-series datasets, and building production-ready machine learning systems. Through these projects, I have gained extensive experience working with imperfect data, identifying artifacts, validating results, and evaluating model performance using rigorous analytical methods. I am particularly interested in applying these skills to physiological waveform data and clinical machine learning applications.

Beyond technical expertise, I enjoy working in multidisciplinary environments where researchers, engineers, and domain experts collaborate to solve challenging problems. My experiences in both academic research and industry have strengthened my ability to communicate technical findings clearly, take ownership of projects, and quickly learn new methodologies when needed.

The combination of cutting-edge machine learning research, clinical impact, and close collaboration with leading researchers across Harvard Medical School, Mass General Brigham, and the Broad Institute makes this opportunity especially exciting. I would welcome the chance to contribute my experience in machine learning, time-series modeling, and data-driven research to your team.

Thank you for your time and consideration. I look forward to the opportunity to discuss my background and learn more about your research initiatives.

Sincerely,

Malhar Gudekar Master of Science in Information Management University of Illinois Urbana-Champaign